

# Hardware Business Management System

## Purpose of This Document

Defines context flow, level 0 data flow, and major system process flows.

## Known Project Scope

- Mobile application and web administration system.
- Super Admin and Shop Admin roles.
- Products, purchases, sales, inventory and stock movement.
- Expenses, capital, debts, profit reporting, and 60/40 allocation support.
- Daily opening and daily closing workflows.
- Offline-first mobile storage using SQLite and central MySQL synchronization.
- Notifications, reports, audit trail, validation, error handling, security, and acceptance criteria.

## Status Label Convention

Confirmed means the item is part of the agreed scope. Proposed means it is a recommended implementation detail that can still change. Pending Decision means the business owner must decide before a developer or Cloud AI treats it as final.

| Status           | Meaning                                                     | How to Treat It                                       |
|------------------|-------------------------------------------------------------|-------------------------------------------------------|
| Confirmed        | Agreed project requirement or design direction.             | Use as baseline for planning and implementation.      |
| Proposed         | Recommended approach that may still change.                 | Validate during design and owner review.              |
| Pending Decision | Business, accounting, or technical rule not fully approved. | Do not hard code. Ask the owner or make configurable. |

## Context Diagram

External actors are Super Admin, Shop Admin, Customer or Supplier where used, and Cloud AI or developer during build.

The Hardware Business Management System receives transaction data, configuration changes, and report requests, then returns saved records, reports, alerts, and sync status.

# Level 0 Data Flow

| Process         | Input                                                 | Output                                 | Data Store                |
|-----------------|-------------------------------------------------------|----------------------------------------|---------------------------|
| Authentication  | Credentials                                           | Session and permissions                | Users Roles Permissions   |
| Shop Operations | Sales purchases expenses debts capital stock activity | Saved transactions and stock movements | SQLite and MySQL          |
| Synchronization | Pending local records                                 | Accepted rejected conflict statuses    | Sync Queue                |
| Reporting       | Filters and report request                            | Reports and exports                    | MySQL report tables/views |
| Audit           | Critical user/system actions                          | Audit trail records                    | Audit Log                 |

## System Flow Diagrams Text

- Sale flow: select product -> validate stock -> save sale -> reduce stock -> audit -> sync if offline -> report update.
- Purchase flow: select product -> enter cost and quantity -> save purchase -> increase stock -> audit -> report update.
- Daily closing flow: open day -> record transactions -> review totals -> resolve exceptions -> close day -> audit.
- Sync flow: pending local record -> send to backend -> validate -> save to MySQL -> return server id -> mark local record synced.

## Open Decisions That Must Not Be Invented

| Decision Area             | What Must Be Confirmed                                                          | Status           |
|---------------------------|---------------------------------------------------------------------------------|------------------|
| Profit formula            | Exact formula for gross profit, net profit, cash profit, and expense treatment. | Pending Decision |
| Stock costing method      | Latest cost, weighted average, FIFO, or another method for cost of goods sold.  | Pending Decision |
| 60/40 destination labels  | Meaning of 60 percent and 40 percent amounts in business reports.               | Pending Decision |
| Allocation timing         | Daily, weekly, monthly, per closing, or owner-approved timing.                  | Pending Decision |
| Tax or VAT handling       | Whether tax calculation is part of first release and how it affects reports.    | Pending Decision |
| Returns refunds discounts | How returns, damaged goods, refunds, and discounts affect stock and profit.     | Pending Decision |

## Acceptance of This Document

- The document is acceptable when the owner can identify which items are confirmed, proposed, and pending decision.
- A developer or Cloud AI can use the confirmed parts without needing to guess the project direction.
- Pending accounting rules remain visible and are not presented as final business rules.